

Def. Let F be a field. Given a polynomial $f(x) \in F[x]$ over F , the Extension $E \stackrel{\text{def}}{=} F(\alpha_1, \dots, \alpha_n)$ where $\alpha_1, \dots, \alpha_n \in \bar{F}$ are roots of f .

is called the splitting field of $f(x)$.

Further, the collection of polynomials $\{f_i(x) \mid i \in I\}$, the intersection of splitting fields of f_i is also called the splitting field.

Thm. Let $\varphi: F \rightarrow F'$ be an isomorphism of fields.

Given $f(x) = a_n x^n + \dots + a_1 x + a_0 \in F[x]$, let $f'(x) = \varphi(a_n) x^n + \dots + \varphi(a_1) x + \varphi(a_0) \in F'[x]$.

Let E be the splitting field for $f(x)$ over F ,

and E' be the splitting field for $f'(x)$ over F' .

Then, φ can be extended on E to E' , isomorphically.

Dummit. sec 13.4.

Prop. Let K be a finite extension of F .

K/F is splitting field

if and only if

every irreducible polynomial in $F[x]$ that has a root in K splits over K .

Proof. Suppose that K is the splitting field of $\{f_i(x) \mid i \in \{1, 2, \dots, n\}, f_i \in F[x]\}$. Since K/F is finite, the collection $\{f_i\}$ was chosen finite. Simply, we say that K is a splitting field of $f = f_1 \cdots f_n \in F[x]$. $\leftarrow$ 약 필요한 가정은 아님.

Let $p(x) \in F[x]$ be an irreducible polynomial such that $p(\alpha) = 0$ for some $\alpha \in K$. Then, for any other root $\beta \in F$ of $p(x)$, there is a conjugate isomorphism $\psi: F(\alpha) \rightarrow F(\beta)$ which fixes F and maps α to β .

Now, by the Extension Theorem, there is an extension of ψ , $\bar{\psi}: K \rightarrow \bar{F}$ such that $\bar{\psi}|_{F(\alpha)} = \psi$. But, since K/F is splitting field and $\bar{\psi}$ fixes F , so $\bar{\psi}[K] = K$. Therefore, $F(\beta) \subseteq K$, so $\beta \in K$.

Conversely, suppose that the below. Since K/F is finite, we can write $K = F(\alpha_1, \dots, \alpha_r)$. Now, let $f_i \in F[x]$ be a minimal polynomial of α_i . Let L be the splitting field of $f_1, \dots, f_r$ over F . By the assumption, all f_i split over K , so $L \subseteq K$. Conversely, each α_i is a root of f_i , so $K = F(\alpha_1, \dots, \alpha_r) \subseteq L$. Consequently, $K = L$. $\square$

Failed try: $\Rightarrow$

Let $f(x) \in F[x]$ be an irreducible polynomial such that for some $\alpha \in K$, $f(\alpha) = 0$. If $\alpha \in F$, then f is just degree 1 polynomial, so there is nothing to prove. If $\alpha \notin F$, then α is a root of for some f_3 , and since f_3 splits over K , so the irreducible f is also minimal polynomial of α , so f divides f_3 , therefore f splits over K .

α 가 어떤 f_3 의 근일 필요는 없다!

Ex) $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ 은 $(x^2 - 2)(x^2 - 3)$ 의 splitting field.
하지만 $\sqrt{2} + \sqrt{3}$ 의 minimal polynomial은 $x^4 - 10x^2 + 1$.

Prop. Let K_1, K_2 be finite extensions of F contained in the field K , and suppose that K_1 and K_2 both are splitting fields over F .

Then, $K_1 K_2$ and $K_1 \cap K_2$ both are splitting fields over F .

Proof. Let K_1 and K_2 be splitting field of f_1 and f_2 , respectively.

Then, for all roots $\alpha_1, \dots, \alpha_k$ and $\beta_1, \dots, \beta_r$ of f_1 and f_2 , respectively,

we can write $K_1 = F(\alpha_1, \dots, \alpha_k)$ and $K_2 = F(\beta_1, \dots, \beta_r)$

Then, the Composition field $K_1 K_2 = F(\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_r)$.

Meanwhile, let L be the splitting field of $f_1 \cdot f_2$. Then,

L contains all roots of $f_1 f_2$, so $K_1 K_2 \subseteq L$. And L is splitting field of $f_1 f_2$, so $L \subseteq K_1 K_2$, so $L = K_1 K_2$ is splitting field.

Let $\alpha \in K_1 \cap K_2 \setminus F$ be given. Then there is minimal irreducible polynomial $f(x)$ over F having α as a root. Since $\alpha \in K_1$ and $\alpha \in K_2$,

the $f(x)$ splits in both K_1 and K_2 , so for $\{\alpha_1, \dots, \alpha_n\}$, the set of all roots of f , $F(\alpha_1, \dots, \alpha_n) \subseteq K_1 \cap K_2$, so f splits over $K_1 \cap K_2$.

Since α was arbitrary, the $K_1 \cap K_2$ is splitting field.

Def. A field K/F is called separable if:

every element of K is the root of a separable polynomial over F .

Thm 13.5

Prop. Let F be any field, and $d, n \in \mathbb{N}$. Then

d divides n if and only if $x^d - 1$ divides $x^n - 1$.

Proof. Suppose that $n = dk$ for some $k \geq 2$. (If $k=1$, we are done)

$$x^n - 1 = x^{dk} - 1 = (x^d)^k - 1 = (x^d - 1)(x^{d(k-1)} + x^{d(k-2)} + \dots + 1)$$

So, $x^d - 1$ divides $x^n - 1$. Conversely, suppose that $x^d - 1$ divides $x^n - 1$.

By the division algorithm, $n = qd + r$ for some $q, r \in \mathbb{Z}$ and $0 \leq r < d$.

$$\text{So, } x^n - 1 = x^{qd+r} - x^r + x^r - 1 = x^r(x^{qd} - 1) + x^r - 1. \quad (q \neq 0, \text{ since } n \geq d)$$

Since $x^{qd} - 1 = (x^d - 1)(x^{d(q-1)} + \dots + x + 1)$ is divided by $x^d - 1$,
by the assumption, $x^r - 1$ is divided by $x^d - 1$. But, since $r < d$,
the remainder r must be zero, so $n = qd$. $\square$

Prop. $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n}$ if and only if d divides n .

Proof. Since

$$\mathbb{F}_{p^k} = \{x \in \overline{\mathbb{F}_p} \mid x^{p^k} - x = 0\}$$

and the fact that

$$\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n} \Leftrightarrow x^{p^d} - x \mid x^{p^n} - x$$

$$\Leftrightarrow p^d - 1 \mid p^n - 1$$

$$\Leftrightarrow d \mid n$$

So proved.

